L2.6 The Gaussian elimination method

Transcript Language

English

Hello and welcome to the maths 2 component of the online B.Sc. program. In this video

we are going to talk about a process called the Gaussian elimination method. Using this

method, we will be able to do a lot of things. So, in particular, we will be able to find

the solutions to any system of linear equations, in a very fixed algorithmic manner. And we can also use it to find the determinant of a square matrix which we kind of saw last

time. And we can also use it to find the inverse of a square matrix which is invertible. So,

let us quickly recall, what was our, what we have seen before.

So, we have seen the following methods to find the solutions of a system of linear equations,

Ax equals b. Namely, we have seen if A is invertible, then the solution is unique. And

it is given by A inverse b. And we can find the solution by either Cramer's rule, that

was the first method we saw. And the other method we saw was to actually compute A inverse.

To do this, we find the adjugate matrix and then we find A inverse b.

So, these are slightly cumbersome methods, because they involve a lot of determinant

calculations. And finding determinants is a very resource intensive process. So, you

have to do a lot of computation to find determinants. And here, we have to find not just one, but

several determinants, and that is how these processes work. So, compared to this, remember that if A is in what we call the reduced row echelon form, then we can find the solution

of, of such a system very easily.

So, if you, if Ax is equal equals b is the system that we are solving, and if A is in

reduced echelon form, then we have the following procedure to find the solutions and note that,

there can be many solutions. Now, because it A is no longer a square matrix, it could

be a rectangular matrix. So, you find the dependent variables. So, these are the ones

which correspond to the leading entries. So, the there is a 1 in the nonzero rows, which

is, which appears at the beginning of the row, the first entry of the, first nonzero

entry of that row looked at from the left, that is a leading term.

So, the column corresponding to that, that column, the corresponding variable. So, if

the ith column is a leading entry, then xi is an, is a dependent variable. And if the

ith column does not have a leading entry, that, that corresponding variable is an independent variable. So, all those columns which do not have leading ones, those correspond to independent

variables. And then how do we find our solutions, you plug in any value that you want for the

independent variables, and you back calculate to find the values of the dependent variable.

And note that each dependent variable will appears in a unique equation that is the advantage

of the reduced row echelon form. And you can use that to back calculate and find the value

of a dependent variable. And then all such values for every choice of values of the independent

variables, we get a solution of our system Ax equals b. Of course, you have to when before

we do this, remember, this is very important that if there is a 0 row in your reduced row

echelon form, and the corresponding entry in b is nonzero, then this system cannot have

a solution.

This was one of the things we observed and we will use this as you go ahead. So, this

tells you all the possible situations. So, it tells you whether there is a solution, if there is a solution, it enlists all of them. And we are going to make use of this

technique. So, we saw this two videos ago and we are going to make use of technique

along with what we saw in the previous video where we took an arbitrary matrix and we used

row operations. So, elementary row operations to reduce it to reduced row echelon form.

So, we are going to put both these two together. And we are going to solve an arbitrary system

of equations Ax equals b

So, what are the contents of this video, we will study what is called the augmented system,

augmented matrix for a system of linear equations, we will study the Gaussian elimination method

to determine all solutions of a system of linear equations. And here, our system may

not have the same number of variables as equations. So, in other words, your A could be a rectangular

matrix, it need not be a square matrix. And finally, we will use this also, this is very

brief to compute the inverse. So, we will use this technique also to compute the inverse of a, of an invertible matrix, of course, which is square.

So, what is the augmented matrix? So, let Ax equals b be a linear system of equations.

So, A is an m by n matrix. So, you have m rows and n columns, so remember that the n

columns correspond to the n variables, and the m rows correspond to the m equations.

And b is as a result, the m by one column vector, or column matrix of constants, which

appears to the right. So, any such equation, we have these three matrices, Ax and b, A

is m by n, x is a column vector a column matrix of size n by 1 and b is a column vector of

size m by 1.

So, we form what is called the augmented matrix of the system. So, this is a new matrix, and

it is of size m by n plus 1. So, there is a small typo here, namely that our n plus

1 should be in the brackets. So, m by n plus 1. So, the first n columns are the columns

of A, and the last column is b. So, this is, this is how I define the augmented matrix of this system. And we have a notation for the augmented matrix, it is A to slash b.

So, this is to remember that the first n columns are coming from A and the last column is coming

from b, can we put this vertical line, so this is just notation. And how are we going

to write this augmented matrix? So, we will write this augmented matrix as follows. So,

if my coefficients for the equation are a 1 1 a 1 2 etcetera. So, then I have a 1 1

a 1 2 all the way up to a 1 n, I put this vertical line here, a 2 1 a 2 2 upto a 2 n.

And then I have a m 1, a m 2 up to a m n.

So now, this is my matrix A remember, so, A came in the first n columns, that is what

the definition says. And in the last column, I have my vector b1, b2, bm. So, this is the

augmented matrix for the system. So, it is a matrix with m rows, same number of rows

as number of equations, and n plus 1 columns, the n columns coming from the coefficients

and the last column coming from the constants. So, what do we do with this augmented matrix?

So, for the augmented matrix, let us, let us first do an example maybe, of an augmented

matrix. So, here is my system of equations, 3x 1 plus 2x 2 plus x 3x 4, 6 x 1, 2 x 2 x

2, 7x 2 plus x 3x 4 is 8. So, A is the matrix 3 2 1 1; 1 1 0 0; 0 7 1 1. You might remember

this from the previous video. So, we have already done the operation of row reduction

on this matrix. And b is the column vector of the constants, which are appearing on the

right, so that is 6 2 and 8.

So, what is the augmented matrix? So, you just take these two, and you put them together,

but you put a line in the middle. So, this line is just to be, it is not sort of part

of the matrix, but it is something that you put to remember that on one side, you have

coefficients and on the other side, you have the constants.

So, what do we do with the augmented matrix? So, this is what the Gaussian, where the Gaussian

elimination method starts. So, you have a system of linear equations, Ax equals b, and

Gaussian elimination is going to tell you how to get all the solutions of this. So,

it will use these two things. First, it will use the augmented matrix. And second, it will

use the technique of reducing to the row echelon form that we have reduced row echelon form

that we studied in the previous video.

And then it will, it will use that for, for matrix and reduced row echelon form. If I

have something like Rx equals c, then I can find all the solutions. This is exactly the

technique. So, let us go through the steps. So, form the augmented matrix. So, we just

saw what that was. Perform the same operations on this augmented matrix, that were used to

bring A into reduced row echelon form.

So, we know how to bring any matrix into reduced row echelon form. So, if you take the matrix

A, there is a series of steps and it is an algorithm, using which you can put it into

reduced row echelon form, so you have to use elementary row transfer, row operations. So, you use the same operations on this new matrix that you used on A in order to put a into

reduced row echelon form.

So, note that this new matrix that you have, need not be in reduced row echelon form. And

indeed, that is the point, if it is not in reduced row echelon form at the end, then

something special is going to happen. And if it is in reduced row echelon form at the

end, then something else is going to happen. So, let R be the sub matrix of the obtained

matrix of the first n columns. So, you look at the so after you do this process on A slash

b, so the augmented matrix, you apply all these elementary row operations.

So, in the end, you will get another new matrix, which is again with m rows and n plus 1 columns.

So, you look at the first n columns, and that matrix you call R. And let c be the sub matrix

of the obtain matrix consisting of the last column. So, the last column consists of c.

And to keep track again of the first ten and the last column, we will put a vertical line between them. So, we write the obtained matrix as R vertical line c.

And now this is the important part, notice that R is in the reduced row echelon form,

it must be, because you took A and you applied all the row operations on A, which we used

on A slash b, that we used to put A into reduced row echelon form. So indeed, R is exactly

the matrix that you got, after you applied all those operations to reduce it into reduced

row echelon form. So, R is exactly the, that matrix.

So, what is next? So the, this is the main point, this is the crucial point, which is

why the entire method works, these solutions of Ax equals b are precisely the solutions

of Rx equals c. So, what have we done, we have taken an arbitrary system of linear equations

Ax equals b, we have performed some operations on this, on the matrices occurring there,

we have found two new matrices R, which is of size m by n and c, which is of size m by

1, where R is reduced row echelon form. And when we take the corresponding system of linear equations Rx equals c, the set of solutions of Rx equals c is exactly the same as the

set of solutions of a Ax equals b.

Now, well, if you think a little about what is happening, you can see that we are already

done here. So, from here, I can finish off how to find the solutions Ax equals b.

Let us list out the procedure entirely, form the corresponding system of linear equations

Rx equals c, find all the solutions of Rx equals c and hence of Ax equals b. So, if

I, as we observed, the solutions of Ax equals b are exactly the same as the solutions of

Rx equals c. That was the punchline of the previous slide. And now so if we find all

the solutions of Rx equals c, then I know all the solutions of Ax equals b.

And how do I find the solutions of Rx equal c? Well, R is in reduced row echelon form.

So, I have a concrete procedure to find these. So, you find the independent variables and

the dependent variables and then you put the independent variables to arbitrary values, and then you back calculate to find the values of the dependent variables. We have done examples

of this as well. We recalled this right at the beginning of this video. So, we have a

very concrete procedure now to, to find the solutions of Ax equals b.

So, let us do an example. So, you might find this, this is the same example that we saw

earlier, where we put the system of linear equations into the, we wrote down the augmented

matrix of the system of linear equations. So, now let us perform the Gaussian elimination

method on that same example. Now, if you look at this matrix on the left, just the matrix,

just this part, the A part.

So, if you look at that A part, then that A part is exactly the matrix that we used

in the previous video, which we use the which we completely computed, what was the reduced

row echelon form for. And so you have to use those same steps over here. So, we can do,

we will do a fresh, of course. So, we will do that a fresh, so the first step is, you get a 1 in the 1 1 place, because you start with the first column, and then you try to

get a 1 in that 1 1 place.

So, you do R 1 by 3. And then you do R2 to minus R1, to cancel all the 0s below. So,

this is what you get. And now notice what is happening here, we are not performing these

operations only on the A part, we are performing it on the entire matrix. So, in that includes

the b part, it includes the column b. So, when we went from the first step, in the first

step, the R1 by 3 was also applied on the column b. So, you have to apply it on the

entire row, not just the row of A but the entire row.

In the second step, we did R2 minus R1. And in R2 minus R1, we got a 0 in the second place

of the last column. So, your performance on the entire second row, not just the A part

of the matrix. Let us continue. So, the next thing we do is we multiply the second row

by 3 in order to bring a 1 over there. Then we cancel out, we cancel out the 0 below that

1, by doing R3 minus 7 R2. Keep noticing what is happening to the last column, you are playing

all this on the last column as well. And then finally, we have to get a 1 in the third row

now. So, we divide by 8. And notice that the 8 in the last column, also got divided by

8, and we got a 1.

So, let us write down this matrix again. And maybe now I will finish this calculation.

So, this is already, the matrix on the left is already in row echelon form. So, now I

am going to reduce it to the reduced row echelon form. So, in order to do that, what do I do?

So, I do R2 plus R3, and R1 minus one third R3. So, what do I get? So, this is still 1

0 0 two third 1 0 then 0 0 1. And, well, we have slightly lucky, we also get a 0 0 1 here.

And then let us see what happens here.

So, R2 plus R3 means I get a 1 in this place. So, I already have a 1 here. And then R1 minus

one third R3. So, 2 minus 1 third. So, that is 6 by 3 minus 1 third. So, that is 5 by

3. And in the next step, what do I get? I do the same thing for the second column. So,

I do R1 minus 2 third R2. So, if I do that, I get 1 0 0 0 1 0 0 0 1 0 0 1. But something

may change here. So, R1 minus two third R2 means you would have five third minus two

third, so five third minus two third is 3 third, which is one, so 1 1 1.

And this is my final matrix. So, in terms of our notation, this is what I call R, and

this is what I call c. Now let us write down this corresponding equation. So, we have Rx

equals c. And now how do I find the solutions for this. So, we look at the independent and

the dependent variables. So, here there is three leading ones, namely column one, column

two, and column three. So, x1 x2 x3 are independent, sorry dependent variables.

And x4 is independent. So, let us put an arbitrary value for x4. So, substitute x4 equals c.

And now let us see what happens to the others. So, x1, actually, there is no choice, x1 is

1, x2, again, there is no choice x2 is 1 if you write down the equation, and in the third equation, we have x3 plus x4 equals 1, but we have put x4 equals c, that means x3 equals

1 minus c. So, the set of solutions of Rx equals c, and hence, Ax equals b is vectors

of the form, so that is x1.

So, you can either write it as a vector or in or explicitly, maybe let me write it explicitly.

So, x1 is 1, x2 is 1, x3 is 1 minus c and x4 is c. And where c can take any choice,

we can have any number of c, so any real number, I can write the same set as in terms of a

vector as 1 1 1 minus c and c, where c belongs to R. So, we have determined all the solutions

of the system of equations, Ax equals b, where A was exactly that matrix that we started

with, in our first example.

So, maybe let us do another example. So, here we have x1 plus x2 plus x3 is 2, x2 minus

3 x3 is 1, 2 x1 plus x2 plus 5 x 3 is 0. So, the matrix representation, so A is a square

matrix, 1 1 1, 0 1 minus 3, 2 1 5, and b is 2 1 0. So, the augmented matrix, you put A and b beside each other, and put a vertical line across them. So, it is three rows and

four columns, where the fourth column is b, and the first three rows are A.

So, let us do this example, maybe explicitly. So, maybe let us write down the first step.

So, this is R3 minus 2 R1, why R3 minus 2R1, well, there is already a 1 in the 1 1-th position.

So, R3 minus 2 R1 cancels at 2 at the, in the first column. And note that we have also

applied this on the last column. So, the last column entry becomes minus 4. So, let us do

one more step. So, you have R3 plus R2.

Now, because once again, our so we want a 1 in the second column, in this part of the

matrix over here, which we already have. So, so I do not have to do anything to make that

1. And now I want to use that 1, and clear off the minus 1 below that. So, to do that,

we do R3 plus R2. And again, the entry in the third, the third entry here changes. So,

let us continue and finish off this example. So, well, this is actually already in row echelon form.

So, now let us put it into reduced row echelon form, to put it into reduced row echelon form,

what do I have to do, I have to get rid of this 1 over here. So, let us do that, to do

that I have to do R2, sorry, R1 minus R2. So, I have 1 0 0 and then 0 1 0 and then 1

minus, minus 3, that is 4, and then minus 3 and 0, and then 1 minus 1, which is 0. So,

0 1 and minus 3, we could actually have skipped this step but I am doing it just because I

want to stick to the procedure.

So, this is in reduced row echelon form. So, this is our R, this is our c. So, let us write

down Rx equals c. So well, so R is is reduced row echelon form. So, I can read off all the

solutions. And now you have to be slightly careful and carefully observe the following.

Look at this last thing over here. So, you have a 0 row in the, on the left meaning in

R, you have a 0 row, but the corresponding constant is not 0.

So, as soon as that happens, what do we know? We know that this does not have solutions,

meaning there is no solution to this equation, maybe that is how I should write it. So, there

is no solution to this system. So, this system does not have solutions. And because this

system does not have solutions, neither does Ax equals b. So, therefore, Ax equals b does

not have solutions. And of course, what does Ax equals b, A is this matrix here 1 1 1 0

1 minus 3 2 1 5, and b was the column 2 1 0. Now, we wrote down the explicit system

in the previous slide. So, we have, we have shown an example where we can use this method

to also check when a system does not have any solution.

So, I hope both these examples are, have cleared up any residual doubts, I will just make a

comment before I move ahead, I need not have done this last step. Anytime in this procedure,

when you hit this kind of situation where you have a zero row, and the corresponding

entry in your, in your constants, on the other side of the vertical line is nonzero, we are saying that in your, you have an equation of the form 0 times x1 plus 0 times x2 plus

0 times x n is equal to something nonzero.

Of course, this cannot have a solution. And you can stop right there and say this does

not have any solutions. But if you feel, sort of uncomfortable with stopping in the middle,

and especially if you want to program this into a computer, then you might as well finish

the entire method. So, this is the power of the Gaussian elimination method, you can find

out all the solutions of a system of linear equations.

So, let us study a particular case, we have studied this in a previous video, namely the

homogeneous system of, system of linear equations. That means Ax equals 0, equals 0. So, remember

that for x equals 0, we always have a solution, namely, x equals 0. If you put each of your

variables to be 0, then certainly this system is solved. So, 0 is always a solution. So,

you cannot have a situation where there is no solution for such a system.

So, homogeneous system always has a solution. So this is called the trivial solution, the

solution Ax equals 0 is called the trivial solution. And of course, that also means that

the other solutions, so if you have a solution, which is not 0, meaning there is some xi,

which is not 0, then that is called a non trivial solution. So, for a homogeneous system,

there are only two different possibilities. One is that the 0 is the unique solution.

So, remember that 0 is always a solution, whatever happens. So, if there is a unique

solution, then it must be 0. So, 0 is a unique solution. That is one possibility.

The other possibility is that there are infinitely many solutions other than 0. And how do we,

how do we get this? Remember that if, if you have

more than one solution, then you have infinitely many solutions. And why is that? That is because

if you have more than one solution, then the only way to get that, that situation is if

one of your variables is a, is an independent variable.

And as soon as you pick up an independent variable, you have an infinitely, you have

infinitely many solutions. That is what they say. The other way of thinking about this

is that suppose you have a solution, let us say, so you have Ax equals 0 and x1 equals

let us say w1, x2 equals w2, and so on xn equals wn is a solution, then so is x1 equals

t times w1, x2 equals t times w2, and xn equals t times wn.

Because if you substitute x1, to be t times w1 etc., in A times x, then what will happen

is the left hand side will be t times A 1 1, let us take the first equation, so, I have

A 1 1 x1 plus A1 2 x2 plus A 1 n xn is 0. And I know that A 1 1 w1 plus A12 w2 plus

A1n wn is indeed 0. But now, if you substitute t times w1 times w2 and so on in that equation,

then you will have A11 times t times w1 plus A1 1 times t times w2 plus A1 n times t times

wn and you can pull out the t common and then you will get t times A11 w1 plus A12 w2 plus

A1n wn, which we know is 0.

So, that is t times 0, which is 0. So, that is, that is the other way of thinking about

this. So, if you have a homogeneous system, that is something special. So, in a homogeneous

system of equations, if there are more variables than equations, then it is guaranteed to have

non trivial solutions, why is this? So, what is what is being said here, so, you have n

variables remember and m equations. So, you have a matrix of size m by n, where these

rows are corresponding to equations and columns are corresponding to variables.

So, now, if you have more variables than equations, then what happens, then when you put it into

reduced row echelon form, when you have any, only m rows, but you have n columns, and n

this strictly greater than m, so, the leading coefficients, the leading ones can occur only

in at most in many places. So, there will be some columns, which do not have leading

ones, that means, there are some independent variables. And as soon as you have independent variables, you will have an infinite number of solutions. So, in particular, you will

have nonzero solutions or non trivial solutions.

In this video, you have studied the Gaussian elimination method, where we put together

the fact that we can find the solutions for a matrix in reduced row echelon form, the

coefficient matrix is reduced row echelon form, along with the fact that any matrix

can be reduced to reduces row echelon form, using elementary row operations. Those were

the contents of the last two videos. And this is a much more powerful method, it is far

more efficient than the previous methods, even when your matrix A is invertible.

So, this is a, this is a method that you will actually use in practice, and it is much more

efficient. And even to compute the inverse of A rather than computing all the miners

and so on this method is far more powerful also to compute the determinant which we did

in the previous video. So, Gaussian elimination method is the most powerful method we know in these contexts. And we have seen explicitly using it, how to find all the solutions, we

can check whether there is a solution at all and then we can take which, once we know there

is at least one which of those are solutions.

And finally, we have seen for homogeneous equations. That is for a homogeneous system,

0 is always a solution. So, the question is, are there other solutions? And we also know

how to find those, again using Gaussian elimination method. Thank you.